

M1 cover note

*M1 is submitted as **two files**: `M1_replication_deck.pdf` and a zip archive of the group’s GitHub repository. This cover note belongs on the **final slide of the deck**, and the same text ships as `COVER-NOTE.md` in the **root of the repository zip**. It is split out here only so the exemplar’s disclosures can be read on their own, and `M1_code.html` ships with the exemplar for the same reason — neither is a third deliverable.*

The syllabus requires a cover note on M0, M1, M2 and M4 containing an **AI Use Statement**, a **per-member contribution note**, and any **external-feedback disclosure**. Two further disclosures belong here whenever they apply: **code derived from the original authors**, and **cross-group conversations that shaped an analytic decision**. This note is short on purpose — it is a disclosure, not an essay.

1. Derivation from the original authors’ code

The article’s data availability statement points to the public NHANES repository and does **not** link analysis code, and we found no public repository for it. **No part of our pipeline is derived from the authors’ code**. The cohort definition, the FLI formula, the MASLD cardiometabolic criteria and the phenotype cut-points are implemented from the Methods text, and every place where that text is ambiguous is logged as a discrepancy on Slide 2 rather than resolved silently.

Reused public code is attributed in the repository README: the Bedogni FLI coefficients, the CKD-EPI 2021 eGFR equation, and the survey-weighted Cox and domain-analysis idioms from the `survey` package documentation.

2. AI Use Statement

AI assistance was used, and it is disclosed here at the level the syllabus asks for — which code was AI-generated and which was human-audited.

Component	AI-generated	Human-audited
NHANES download and merge (<code>01_download.R</code> , <code>02_build_analytic.R</code>)	Drafted with AI	Every variable derivation checked against the NHANES codebook
FLI, MASLD criteria, phenotype groups	Drafted with AI	Checked line by line against the Methods and against Bedogni (2006)
P1–P5 increment code	Drafted with AI	Re-executed; every reported figure regenerated from the executed file
<code>M1_code.qmd</code> assembly and checks	Drafted with AI	Check logic reviewed; deliberately written to fail rather than warn
Slide text and interpretation	Human-owned interpretation; AI-assisted drafting and wording	Every sentence read, edited and verified; the judgments on Slides 7–8 are ours

Two things we want on the record, because they are the point of the auditing requirement:

- **AI-written code was wrong in a way that mattered, and the audit caught it.** The alcohol-exclusion rule applied `ALQ130 > 2` to raw values, which silently counted NHANES reserved codes 777 (“refused”) and 999 (“don’t know”) as drink counts. That reclassified nine refusals as heavy drinkers and moved the excluded count from 2,063 to 2,072. It was found because the deck is assembled from the executed code rather than typed beside it; it is fixed, and the fix is visible in `P1_code.qmd`.

- We do not offer “the AI made a mistake” for anything on these slides. Under the No-Excuse Clause, every figure here is ours to defend at M3.

3. Per-member contribution note

This is the Instructor exemplar, so it has no group members. A real submission replaces this section with one line per member, in the syllabus’s format:

AB: cohort reconstruction, survey weighting code. CD: DAG, confounder set, Table 1. EF: slides, missing-data sensitivity analysis.

For the exemplar itself: written by the course instructor, with the AI assistance disclosed in section 2.

4. External feedback

Yes — external feedback was received, and it changed the submission. This exemplar was put through several rounds of adversarial review by independent large language models, which were given the artifacts and asked to find errors. That review is the reason for, among other things: the split of the cohort funnel into typed steps; the withdrawal of a claim that the crude sex difference was “confounding, not true modification”; the separation of the full analytic file (6,371) from the locked domain (6,048); and the rebuild of `M1_code.qmd` as an assembly notebook after a reviewer noticed the deck quoted a design-aware interaction that no increment computed.

Being precise about what that review contributed. The adversarial-review models provided critique and, in some cases, **proposed wording that informed or was incorporated into the final text.** The instructor selected, edited, verified, and takes responsibility for the final wording and the scientific judgments. Separate AI coding assistance is disclosed in §2. No external contributor ran an analysis or altered code directly.

Under the External Feedback Policy a student group doing this would need prior instructor approval, and the policy’s “comments and suggestions only” limit is stricter than what happened here — which is exactly why the disclosure says so plainly rather than claiming the narrower thing.

5. Cross-group conversation and review

Not applicable to the exemplar. A real submission states here whether a conversation with another group materially shaped an analytic decision, and names the group. From M2 onward it also states whether reviewing another group’s M1 influenced a change — the change being the group’s own, justified and implemented on its own paper. A reviewed group’s code, text, figures, output and distinctive implementation are never carried across. The syllabus is explicit that disclosing the influence is a strength rather than a confession.

Reproducibility statement

`M1_code.qmd` does not re-implement P1–P5. Each increment writes a result object to `../_results/` containing named, already-formatted **facts**; M1 reads those, adds the facts from its own two analyses, and writes `M1.json` and `paper.json`.

No number is typed onto the slides. The deck cites facts by name — `{{ P4.facts.w_m1_iii }}` — and the build substitutes them. An unresolved or misspelled reference fails the build, as does a stray typed decimal in the deck source. Figures attributed to the article are transcribed once, in the notebook, each with the string that must be found in the paper’s PDF; a failed check fails the render.

What this guarantees is that every number shown was produced by the named field of a named result object. It does **not** guarantee that the surrounding prose describes that number correctly — drift from retyping is prevented structurally, interpretive drift is not. Nor does it guarantee that a result object is current: when

the underlying data are rebuilt, each increment has to be re-run so that it re-emits its facts before the deck is rebuilt, or the deck will faithfully show the old values.

The two analyses M1 runs itself are stated in the notebook: the replication table, and the design-aware interaction that P3 explicitly deferred to M1.